Gluing fibrations along a span: a fibration over the pushout whose total space is again a pushout

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'lem:fibration-over-pushout' (source lines 1235-1259, Lemma 8.5.3); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. No context span is retained: the statement and proof are self-contained in the pinned author file apart from one labelled invocation.

Editorial note. The retained counter values (chapter 8, section 5, theorem offset 2) make the counted lemma print as Lemma 8.5.3, matching the original book numbering.

Editorial note. The single external reference is the flattening lemma of chapter 6 invoked at source line 1273; its number is taken from the label-number dump of the pinned book snapshot.

Editorial scope. One original result of the HoTT Book, counted once: the gluing of fibrations along a span (Lemma 8.5.3).

Allowed standard prerequisites (editorial).

- Pushout of a span. Y sqcup^X Z is generated by inl : Y -> ..., inr : Z -> ... and glue : prd x:X inl(j x) = inr(k x).. It is the base of the fibration E constructed in the proof.
- Flattening lemma. For a family P over a coequalizer defined recursively by a family C and a family of equivalences D, the total space Sigma P is equivalent to the flattened higher inductive type.. It is applied at source line 1273 with the explicit values listed afterwards.
- Univalence. An equivalence between types gives a path between them by ua.. It is what makes map E (glue x) = ua(e_X(x)) a well-typed clause of the recursive definition.

Source reference. Original theorem 6.12.2: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 1780–1780. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Lemma 8.5.3. *Let $\mathscr{D} = (Y \xleftarrow{j} X \xrightarrow{k} Z)$ be a span and assume that we have*

- *Two fibrations $E_Y : Y \to \mathcal{U}$ and $E_Z : Z \to \mathcal{U}$.*
- *An equivalence e_X between $E_Y \circ j : X \to \mathcal{U}$ and $E_Z \circ k : X \to \mathcal{U}$, i.e.*

$$e_X : \prod_{x:X} E_Y(j(x)) \simeq E_Z(k(x)).$$

Then we can construct a fibration $E : Y \sqcup^X Z \to \mathcal{U}$ such that

- *For all $y : Y$, $E(\mathsf{inl}(y)) \equiv E_Y(y)$.*
- *For all $z : Z$, $E(\mathsf{inr}(z)) \equiv E_Z(z)$.*
- *For all $x : X$, $E(\mathsf{glue}(x)) = \mathsf{ua}(e_X(x))$ (note that both sides of the equation are paths in $\mathcal{U}$ from $E_Y(j(x))$ to $E_Z(k(x))$).*

Moreover, the total space of this fibration fits in the following pushout square:

$$\begin{array}{ccc} \sum_{(x:X)} E_Y(j(x)) & \xrightarrow{\mathsf{id}\times e_X}_{\sim} & \sum_{(x:X)} E_Z(k(x)) \xrightarrow{k\times\mathsf{id}} \sum_{(z:Z)} E_Z(z) \\ \downarrow{\scriptstyle j\times\mathsf{id}} & & \downarrow{\scriptstyle \mathsf{inr}} \\ \sum_{(y:Y)} E_Y(y) & \xrightarrow[\mathsf{inl}]{} & \sum_{(t:Y\sqcup^X Z)} E(t) \end{array}$$

Proof. We define E by the recursion principle of the pushout $Y \sqcup^X Z$. For that, we need to specify the value of E on elements of the form $\mathsf{inl}(y)$, $\mathsf{inr}(z)$ and the action of E on paths $\mathsf{glue}(x)$, so we can just choose the following values:

$$\begin{aligned} E(\mathsf{inl}(y)) &:\equiv E_Y(y), \\ E(\mathsf{inr}(z)) &:\equiv E_Z(z), \\ E(\mathsf{glue}(x)) &:= \mathsf{ua}(e_X(x)). \end{aligned}$$

To see that the total space of this fibration is a pushout, we apply the flattening lemma (original theorem 6.12.2) with the following values:

- $A :\equiv Y + Z$, $B :\equiv X$ and $f, g : B \to A$ are defined by $f(x) :\equiv \mathsf{inl}(j(x))$, $g(x) :\equiv \mathsf{inr}(k(x))$,
- the type family $C : A \to \mathcal{U}$ is defined by

$$C(\mathsf{inl}(y)) :\equiv E_Y(y) \qquad \text{and} \qquad C(\mathsf{inr}(z)) :\equiv E_Z(z),$$

- the family of equivalences $D : \prod_{(b:B)} C(f(b)) \simeq C(g(b))$ is defined to be e_X.

The base higher inductive type W in the flattening lemma is equivalent to the pushout $Y \sqcup^X Z$ and the type family $P : Y \sqcup^X Z \to \mathcal{U}$ is equivalent to the E defined above.

Thus the flattening lemma tells us that $\sum_{(t:Y\sqcup^X Z)} E(t)$ is equivalent to the higher inductive type $E^{\mathrm{tot}'}$ with the following generators:

- a function $\mathsf{z} : \sum_{(a:Y+Z)} C(a) \to E^{\mathrm{tot}'}$,
- for each $x : X$ and $t : E_Y(j(x))$, a path $\mathsf{z}(\mathsf{inl}(j(x)), t) = \mathsf{z}(\mathsf{inr}(k(x)), e_X(t))$.

Using the flattening lemma again or a direct computation, it is easy to see that $\sum_{(a:Y+Z)} C(a) \simeq \sum_{(y:Y)} E_Y(y) + \sum_{(z:Z)} E_Z(z)$, hence $E^{\mathrm{tot}\prime}$ is equivalent to the higher inductive type E^{tot} with the following generators:

- a function $\mathsf{inl} : \sum_{(y:Y)} E_Y(y) \to E^{\mathrm{tot}}$,
- a function $\mathsf{inr} : \sum_{(z:Z)} E_Z(z) \to E^{\mathrm{tot}}$,
- for each $(x,t) : \sum_{(x:X)} E_Y(j(x))$ a path $\mathsf{glue}(x,t) : \mathsf{inl}(j(x),t) = \mathsf{inr}(k(x), e_X(t))$.

Thus the total space of E is the pushout of the total spaces of E_Y and E_Z, as required. □